
Algorithm 1 Final Classification

1: **Input:** Feature maps $F_{\text{mel, ECG}}, F_{\text{mel, PCG}} \in \mathbb{R}^{B \times 1 \times 128 \times T}$, Training set $\mathcal{D}_{\text{train}}$

2: **Output:** Predicted class label $\hat{y} \in \{\text{Normal, Abnormal}\}$

3: Initialize network weights $\Theta \leftarrow \text{ConvNeXt-Tiny}(\text{in_ch} = 1)$

4: Set hyperparameters: $\eta = 10^{-5}$ (learning rate), $\lambda = 5 \times 10^{-4}$ (weight decay), $\alpha = 0.05$ (smoothing)

5: **for** $e = 1$ **to** 25 **do**

6: **for** each mini-batch $(X_{\text{ecg}}, X_{\text{pcg}}, Y) \in \mathcal{D}_{\text{train}}$ **do**

7: Extract modality-specific feature representations:

$$Z_{\text{ecg}} = \mathcal{F}_{\theta_1}(X_{\text{ecg}}), \quad Z_{\text{pcg}} = \mathcal{F}_{\theta_2}(X_{\text{pcg}})$$

8: Compute concatenated multimodal feature vector:

$$Z_{\text{fusion}} = Z_{\text{ecg}} \parallel Z_{\text{pcg}}$$

9: Map through classification head to obtain logits:

$$S = W_c Z_{\text{fusion}} + b$$

10: Construct label-smoothed target distribution:

$$Y^{\text{smooth}} = (1 - \alpha)Y + \frac{\alpha}{C}$$

11: Evaluate Cross-Entropy optimization objective:

$$\mathcal{L}(\Theta) = - \sum_{c=0}^{C-1} Y_c^{\text{smooth}} \log \left(\frac{\exp(S_c)}{\sum_j \exp(S_j)} \right)$$

12: Update parameters via AdamW optimizer:

$$\Theta \leftarrow \Theta - \eta (\nabla_{\Theta} \mathcal{L}(\Theta) + \lambda \Theta)$$

13: **end for**

14: **end for**

15: Compute final prediction probabilities:

$$\hat{P} = \text{Softmax}(S)$$

16: **return** $\hat{y} = \arg \max \hat{P}$
